

It is time to talk about Z

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Abstract

The atomic number Z is so deeply embedded in modern physics and chemistry, and in our understanding of matter, that questioning its foundational status can appear almost sacrilegious. Yet Z was not discovered as a primitive property of nature but introduced historically as an ordering index that later acquired explanatory and ontological authority.

Across a series of large-scale empirical studies of atomic spectral data, we show that simple scaling laws, periodic organization, physical-property classification, atomic radii and heavy-element predictions, spectral bifurcation and convergence patterns all emerge from relationships among only two experimentally measured spectral parameters. The two-variable description organizes nearly one hundred elements, ionization sequences, hydrogen spectral families, conductivity classes, magnetic classes, and the heavy-element K-edge series. Remarkably, these structures become visible when atomic spectra are organized by a directly measured, system-specific spectral boundary ν_{max} , rather than by atomic number.

The resulting boundary-controlled organization, spanning decades in frequency across different scales, does not emerge naturally from atomic-number representation and, in many cases, is obscured by it. The empirical evidence indicates that atomic number does not generate, constrain, or organize the observed spectral structure. Its spectroscopic utility derives solely from preserving the ordering of atomic K-edge frequencies, which constitute one particular realization of the more general spectral-boundary parameter identified here. This degree of empirical predictive compression warrants a reassessment of atomic number as the primary organizing coordinate of atomic spectroscopy.

Keywords: Atomic number | Atomic spectra | Spectral boundary | Spectral structure | Spectral scaling | Spectral framework | Predictive compression | K-edge frequency

I. Historical Context

Around 1870, Mendeleev constructed the periodic table by arranging the known chemical elements according to their atomic weights (relative atomic masses).^{1,2} When the sequence dictated by mass conflicted with established chemical properties, he prioritized the chemical classification. A well-known case is tellurium (≈ 127.6 amu) and iodine (≈ 126.9 amu): despite iodine's lower weight, Mendeleev placed it after tellurium, because its reactivity clearly identified it as a halogen, whereas tellurium belonged with the chalcogens. Although Mendeleev assumed the discrepant atomic weights were probably measured incorrectly, his insistence on preserving chemical periodicity effectively maintained a consistent ordinal sequence of elements. This sequence was not yet understood as a distinct physical quantity, but it provided a stable index of elemental order. In 1913–1914, Moseley's X-ray spectroscopy showed that the square root of the frequency of characteristic X-ray lines increases linearly with an element's ordinal position in the periodic classification, providing direct spectroscopic support for elemental order independently of atomic weight.^{3,4}

Shortly afterward, while working within the Bohr model of the atom, Walther Kossel (1916) introduced the notation Z , from the German *Zahl* ("number"), reinterpreting the existing ordinal index as the charge on the nucleus.⁵ In 1919, Rutherford's interpretation of nitrogen disintegration reinforced the proton model of the observed charge and suggested that hydrogen nuclei are fundamental constituents.⁶ Therefore, what began as a chemically motivated ordering that sometimes required overriding measured weights was ultimately reified as the proton number, becoming a fundamental physical constituent of the atom.

Within the original Moseley's spectral law, Z functioned as an empirical indexing parameter for characteristic X-ray frequencies. Its subsequent incorporation into the Bohr–Kossel framework reinterpreted this ordinal index as nuclear charge.⁷ This treatment mirrors a broader pattern in the history of atomic spectroscopy, where indices initially introduced as empirical ordering devices subsequently acquire deeper physical interpretations. Another example is the use of integer indices n_1 and n_2 in the Rydberg spectral formula as pragmatic ordering parameters that were later reinterpreted as quantum energy levels.⁸

From a spectroscopic standpoint, the use of the integer Z carried a notable tension. Spectroscopic relations are normally formulated in terms of time or frequency-domain variables, yet here an external integer label was applied to spectral data and subsequently reinterpreted as a structural property of that data. This tension was reinforced by the fact that Z is restricted to integer values, whereas the frequency domain is continuous, raising questions about the extent to which Z preserves the underlying organization of atomic spectra, and whether some of the apparent irregularities observed in Z -based representations reflect the spectra themselves or the indexing scheme imposed upon them. Historically, these irregularities were accommodated through successive theoretical refinements, including relativistic corrections, QED effects, shielding adjustments, and transition rules, all of which retained Z as the organizing variable.

Following the identification of Z with nuclear charge, spectroscopic theory largely adopted atomic number as the natural independent variable against which spectral regularities were expressed. Consequently, comparatively little attention was devoted to the possibility that

directly measured spectroscopic quantities might themselves serve as alternative organizing coordinates. This historical trajectory raises a methodological question: to what extent does the choice of Z as the principal coordinate reveal intrinsic spectroscopic structure, and to what extent might it constrain or obscure empirical regularities that become more apparent when spectra are represented in terms of spectrally derived variables?

II. Empirical Assessment of Z -Based Spectral Representations

The methodological questions raised in the preceding historical discussion can now be examined empirically. Recent large-scale analyses of atomic spectral data derived from the NIST Atomic Spectra Database have investigated the organization of atomic spectra using both conventional Z -based representations and alternative descriptions constructed directly from measured spectral observables.⁹⁻¹¹ These studies span multiple levels of atomic structure, from pooled spectral fingerprints and K -edge scaling relations to the frequency-domain recovery of periodic organization. The empirical results provide a basis for evaluating the representational consequences of choosing Z rather than an intrinsic spectral variable as the principal coordinate for organizing atomic spectra, and reveal a major contrast between Z and the spectral coordinate in the organization, compression, generalization, and prediction of atomic spectral properties.

1. Systematic nonlinearity in the global spectral fingerprint

The first indication that coordinate choice influences the representation of atomic spectra appears in the global scaling of pooled atomic spectral frequencies.⁹ When all experimentally measured frequencies from 95 elements are plotted against atomic number Z , the data exhibit pronounced curvature and systematic departures from linearity across the two X-ray spectral branches (K series and L series). In contrast, when the same frequencies are expressed as functions of the intrinsic spectral boundary ν_{max} , defined as the K -edge frequency of each atom, the data collapse onto a small number of nearly linear branches spanning several orders of magnitude (Fig. 1, Ref. 9). This is not a marginal improvement. For the K_α transition, regression under (ν_{max} yields ($R^2 = 0.99999$), whereas the conventional Z -based representation requires piecewise-linear approximations, multiple breakpoints, and series-dependent screening offsets σ to achieve comparable residuals.

The significance of this result lies not simply in the improvement of fit quality, but in what it suggests about the geometry of the underlying data. The same spectral frequencies are being represented in two different coordinate systems. If the observed curvature were an intrinsic property of the frequencies themselves, it would persist regardless of the coordinate used to organize them. Instead, much of the curvature disappears when Z is replaced by the directly measured spectral quantity ν_{max} . This observation suggests that a substantial component of the

nonlinearity traditionally associated with atomic spectral scaling arises from the choice of organizing variable rather than from the spectral data alone.

The contrast is particularly evident in the role of the screening constant σ . In Moseley's formulation, σ was introduced as an empirical offset to restore approximate linearity. In practice, however, its value is not universal, varying systematically between spectral series and requiring adjustment across different regions of the periodic table. Under ν_{max} description, by contrast, no analogous element-specific correction is required to recover a single global scaling relation. What is empirically clear is that ν_{max} recovers the observed scaling with substantially fewer parameters than the corresponding Z -based representation.

More broadly, the emergence of linear spectral bands under ν_{max} raises a methodological question. Why should a single measured spectral quantity organize a large fraction of atomic spectral frequencies into a low-dimensional structure, while representations based on atomic number require piecewise descriptions and additional corrective terms? Regardless of the interpretation, the existence of this band structure indicates that ν_{max} captures a degree of global spectral organization that is only partially visible when the same data are indexed by Z . The reduction of a high-dimensional spectral dataset to a small number of coherent linear manifolds suggests that ν_{max} may function not merely as another correlated variable, but as a coordinate more closely aligned with the intrinsic geometry of atomic spectral organization.

2. Recovery of inter-atomic spectral lineage and branching architecture

The reduction of curvature observed in the previous section raises a further question: does the choice of the organizing coordinate affect only regression quality, or does it alter the amount of structural information recoverable from the spectral data? Examination of the full frequency dataset suggests the latter.

When the complete set of atomic spectral frequencies from 95 elements is normalized by the upper spectral boundary of each element (ν_{max}) and represented as $\log(\nu/\nu_{max})$ versus $\log(\nu_{max})$, a coherent branching architecture emerges across the entire dataset.^{9,10} Individual atoms appear as dense horizontal plateaus, while successive atoms are connected by continuous mode trajectories that undergo discrete splitting events at specific spectral thresholds (Fig.2, Ref. 9 and Fig. 4, Ref. 10). These splitting events generate new branches while preserving the overall spectral envelope, producing a lineage-like organization that is recoverable directly from the frequency data without reference to atomic number, electron configuration, or chemical classification.

A prominent example occurs near the Ne–Na transition, where the spectral organization changes abruptly and a new branch emerges. Beyond this point, the two principal branches evolve along distinct trajectories while remaining embedded within the same global scaling structure. The resulting geometry is not a collection of isolated atomic fingerprints, but a connected organization in which spectral modes can be traced across multiple atoms through successive branching events.

The same organization is not recovered when the identical frequencies are represented using atomic number as the organizing coordinate. Under Z -based normalization, the branching geometry is absent, and the plateau-and-splitting architecture that defines the spectral lineage is not visible. Quantitative analysis supports this visual observation: ν_{max} -based scaling yields improved clustering metrics and complete atomic coverage, whereas Z -based scaling leaves a subset of atoms ungrouped despite using the same underlying spectral data.¹⁰

The previous section showed that ν_{max} linearizes global spectral scaling relationships. Here, the distinction concerns the recovery of organizational structure itself. A coordinate transformation cannot create information that is absent from the data; however, it can determine whether existing structure becomes observable. In this case, the branching architecture is visible when the data are organized by ν_{max} and obscured when they are organized by Z .

This observation raises a broader methodological question. Why should a single measured spectral quantity reveal a connected branching organization spanning 95 atomic spectra, while a coordinate historically adopted as the principal organizing variable of atomic spectroscopy does not? Whatever the ultimate interpretation, the emergence of a reproducible spectral lineage suggests that atomic spectra possess a degree of low-dimensional organization that is more readily expressed in terms of spectral observables than through atomic-number indexing alone. The following section examines whether this same organization extends to the empirical recovery of periodic structure.

3. Spectral origin of periodic organization

The branching architecture identified in the previous section raises a fundamental question: do the observed bifurcation events correspond to any recognized level of atomic organization, or are they merely geometric features of the spectral data? Analysis of the normalized spectral coordinate system indicates that these bifurcations coincide directly with the emergence of periodic structure.¹⁰

When the first ionization frequency is expressed relative to the spectral ceiling through the representation $(\nu_{ion}/\nu_{max}, \nu_{max})$ in logarithmic space, elements organize into a series of bounded spectral bands embedded within a common power-law envelope (Fig. 1, Ref. 10). Statistical clustering identifies approximately seven stable bands corresponding sequentially to Periods 2–7 of the periodic table. The bands are not imposed by chemical classification or atomic-number labels but emerge directly from the spectral data themselves, indicating that periodic segmentation is already encoded within the frequency-domain organization of atoms.

More importantly, the major bifurcation events identified in the spectral lineage correspond precisely to the boundaries between successive periods.¹⁰ The first major split occurs at Neon and marks the emergence of Period 3. Subsequent bifurcations at Potassium, Krypton, and Xenon/Barium generate Periods 4, 5, and 6 respectively (Fig. 4, Ref. 10). Each splitting event introduces a new spectral branch and establishes a new bounded spectral domain within which subsequent atomic differentiation occurs. Period boundaries therefore appear not as externally

imposed classifications but as the direct consequence of discrete bifurcation events in the spectral lineage.

The same spectral framework explains differentiation within periods. While primary bifurcations generate new periods, secondary splitting events occurring within existing spectral domains produce finer subdivisions without creating new spectral ceilings. Local bifurcations near Zn and In, for example, coincide with transitions between transition and post-transition regimes. These secondary splitting events generate the parallel trajectories that later appear as chemical groups, while preserving the larger periodic structure established by the primary bifurcations. Periodic organization therefore emerges hierarchically: major splits generate periods, whereas secondary splits generate group-level differentiation within those periods.

This hierarchical organization becomes particularly evident in the $(\nu_{ion}/\nu_{max}, \nu_{max})$ representation. Within the global power-law envelope, elements belonging to the same chemical group align along approximately parallel power law trajectories with closely related exponents but systematic vertical offsets (Fig. 1 and 2, Ref. 10). Alkali metals, alkaline-earth metals, halogens, noble gases, transition metals, and post-transition metals occupy distinct paths within the same spectral space. Diagonal relationships such as the metalloid sequence and Knight's Move relationships emerge naturally from the geometry, indicating that both periodic recurrence and chemical similarity are encoded within a common spectral framework.

The significance of these observations becomes clearer when compared with atomic-number representations. Atomic number preserves the overall ordering of elements and reproduces coarse periodic segmentation, but the bifurcation structure itself is not visible. The branching events that define the emergence of new periods disappear, leaving only the sequential ordering of elements (Fig. S11, Ref. 10). Statistical analysis further shows increased dispersion and incomplete clustering relative to the (ν_{max}) -based organization (Fig. S10, Ref. 10). Thus, while Z preserves the sequence of elements, it does not recover the generative structure through which periodic differentiation emerges in the spectral data.

These results extend the argument developed in the previous sections. The global spectral fingerprint revealed a low-dimensional scaling structure, and the normalized spectral representation exposed a branching lineage connecting atomic spectra. The present analysis shows that this branching process is not merely geometric: its major bifurcations correspond to the formation of new periods, while its secondary bifurcations generate the trajectories associated with chemical groups. Periodicity therefore appears as a direct manifestation of the branching organization already present within the empirical spectral data, rather than as an independent classification imposed upon them using Z and chemical behavior classification.

4. Explanatory power of the spectral coordinate ν_{max} versus Z : the liquid state of mercury and magnetic properties of gadolinium

The preceding section established that the $(\nu_{ion}/\nu_{max}, \nu_{max})$ representation is not merely descriptive. The bifurcation structure recovered from the spectral data provides a direct account

of how periods and groups emerge. A further question is whether the same coordinate system can explain specific anomalous elemental properties that traditionally require additional theoretical interpretation.

Mercury provides a well-known example. Its liquid state at ambient conditions has long been regarded as anomalous among metals. Conventional explanations invoke a combination of electronic configuration, relativistic effects, and weakened metallic bonding. These explanations are widely accepted, but they require the introduction of theoretical constructs beyond the empirical organization of the elements themselves.

In the spectral coordinate system, the anomaly becomes directly visible, or to be precise, it is not an anomaly anymore. When elements are represented in $(\nu_{\text{ion}}/\nu_{\text{max}}, \nu_{\text{max}})$ space, chemical groups occupy distinct power-law scaling trajectories. Mercury does not lie on the continuation

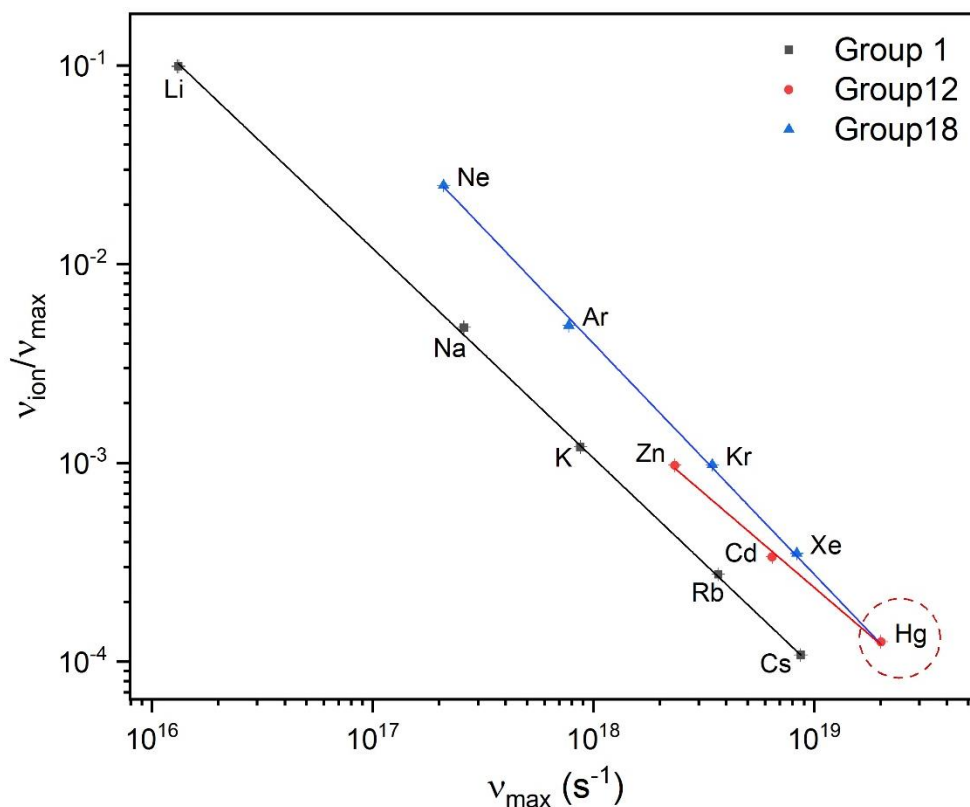


Figure 1. Normalized ionization frequency ($\nu_{\text{ion}}/\nu_{\text{max}}$) plotted against the spectral boundary ν_{max} for representative elements from Groups 1, 12, and 18 on logarithmic axes. The figure illustrates two key features of ν_{max} -based organization. First, the progressive convergence of the group trajectories demonstrates the contraction of periodic spectral differentiation Δ_p with increasing ν_{max} , consistent with the global scaling law $\Delta_p \propto \nu_{\text{max}}^{-1.19 \pm 0.01}$. Second, Hg occupies a position at the intersection of the Group 12 and Group 18 trajectories. This transitional location provides a direct geometric explanation for the anomalous behavior of Hg within Group 12 and illustrates how explanatory structure emerges naturally from spectral organization. Data reproduced from Ref. 10 supplementary material.

of the Group 12 metallic branch. Instead, it is displaced toward the noble-gas trajectory and occupies a position near the intersection between the two scaling regimes.

This observation provides a simple empirical explanation for mercury's unusual behavior. Rather than appearing as an exceptional member of an otherwise regular metallic sequence, Hg occupies an intermediate position within the spectral geometry itself (Fig. 1). Its anomalous properties correspond to a measurable location in the scaling structure, specifically a transition region between metallic and noble-gas-like domains. The explanation is therefore expressed directly in terms of observed spectral relationships rather than through additional theoretical entities.

This interpretation does not reject existing theoretical accounts. Instead, it highlights a difference in explanatory style and in compression. The conventional explanation requires supplementary theoretical constructs to account for mercury's position in the periodic table, whereas the spectral description makes the anomaly visible as a geometric feature of the empirical data. In this sense, the ν_{max} -based spectral framework functions as more than a descriptive coordinate system. It provides a data-driven explanatory map in which anomalous properties emerge naturally from the organization of measured atomic observables.¹⁰

A second example is provided by gadolinium (Gd), whose magnetic behavior is unusual among the lanthanides. Below its Curie temperature (~ 293 K), Gd is ferromagnetic, whereas above this temperature it becomes paramagnetic. Conventional explanations attribute this transition to thermal disruption of long-range spin alignment: below the Curie point, exchange interactions maintain collective magnetic ordering, while above it thermal fluctuations destroy that ordering despite the persistence of local magnetic moments. In the spectral scaling representation, Gd occupies a similarly informative position. As shown in (Fig. 3C, Reference 10) Gd lies directly on the ferromagnetic scaling trajectory defined by Fe, Co, and Ni, while simultaneously residing near the intersection between that trajectory and the branch occupied by the paramagnetic lanthanides. Its dual magnetic character therefore appears geometrically encoded in its location within the spectral space. Rather than appearing as an anomalous lanthanide requiring special treatment, Gd occupies a transition region between two distinct magnetic regimes.

Hg and Gd illustrate a common pattern. In both cases, properties traditionally regarded as exceptional become visible as positional consequences of the spectral geometry. The significance of these examples is that they reveal the anomalies directly within the empirical organization of the data. The spectral coordinate therefore provides explanatory information that is absent from atomic-number indexing alone. While Z preserves elemental order, it does not reveal the branch intersections, regime boundaries, or transition zones through which these behaviors become visible. The explanatory content emerges from the geometry of the spectral organization itself.

5. Predictive power and the emergence of a scale-invariant spectral law

The preceding sections showed that the ν_{max} -based description recovers organizational and explanatory structure that remains invisible or only partially visible when atomic spectra are indexed by atomic number. A more demanding test is whether the same coordinate system

possesses predictive power. Explanation can often be achieved retrospectively; prediction requires the existence of quantitative constraints that limit the range of possible outcomes before the data are known.

A remarkable example emerges in the spacing between successive K-edge frequencies. For heavy elements beyond Xe, the difference between neighboring spectral ceilings obeys a simple linear relation,

$$\Delta\nu_{\max} = a + b\nu_{\max},$$

with $a=(1.801 \pm 0.009) \times 10^{17} \text{ s}^{-1}$, $b = 0.0201 \pm 0.00005$, $R^2 = 0.99981$. The relation spans the experimentally measured range ($Z=55\text{--}99$) and describes the spacing of consecutive K-edge frequencies with near-perfect linearity.¹¹ Most significantly, the fractional spacing $\Delta\nu_{\max}/\nu_{\max}$ approaches a constant asymptotic value of approximately 2.01%, indicating that successive K-edges become separated by a fixed fraction of the local spectral scale rather than by a fixed absolute increment, regardless of Z .

The importance of this result lies in the nature of the variable itself. The quantity $\Delta\nu_{\max}/\nu_{\max}$ is a dimensionless spectral invariant constructed entirely from experimentally measured frequencies. It expresses the spacing of the spectral sequence relative to its local scale and therefore remains meaningful independent of units, calibration conventions, or theoretical assumptions. By contrast, the corresponding quantity $\Delta Z/Z$ has no analogous spectroscopic interpretation. Atomic number provides an ordering index, but the fractional spacing between neighboring integers carries no known physical significance. Consequently, the scale-invariant structure becomes visible only when the sequence is expressed in terms of the spectral coordinate itself.

The predictive consequences are substantial. Starting from a single experimental anchor point (Cs, $Z=55$), recursive application of the spacing law predicts K-edge frequencies throughout the heavy-element sequence with sub-percent accuracy (Fig. 1, Ref. 11). Seven atoms in the Pb–U gap, whose adopted NIST values were not used in fitting, are recovered with a median deviation of approximately 0.1%. Extension to the synthetically confirmed elements ($Z=111\text{--}118$) reproduces independent Dirac–Fock reference values to sub-percent accuracy despite using no relativistic Hamiltonian, QED corrections, shell-structure information, or element-specific parameters. Leave-one-out cross-validation across the experimental dataset yields an RMS prediction error of only 0.020%, confirming that the relation generalizes robustly beyond the specific atoms used to determine its coefficients, including into elements that are not measured or synthesized yet.

The significance of this predictive accuracy extends beyond numerical performance. If K-edge frequencies were independently determined quantities whose values depended primarily on the detailed internal structure of each atom, no single-neighbor recursion would be expected to succeed at this level of precision across a factor of six in frequency. Instead, the observed behavior indicates that successive K-edges are strongly constrained by their position within a larger spectral sequence. The heavy-element regime therefore exhibits a degree of compressibility that is unusual for atomic data: a large collection of measured frequencies can be

represented, to high accuracy, by a single dimensionless spacing parameter of approximately 2%. The same spacing law applies similarly to light elements, using a different power law scaling between $\Delta\nu_{\max}$ and $\nu_{\max}$.

From the perspective developed throughout this work, the spacing law provides a quantitative manifestation of the same low-dimensional spectral organization already visible in the branching and periodic structures. The spectral coordinate does not merely reveal how atomic spectra are arranged; it constrains how they evolve from one atom to the next. The existence of a stable asymptotic spacing ratio implies that neighboring atoms are not spectroscopically independent entities but successive realizations of a common scaling process. This level of predictability is not apparent in atomic-number representations, where the corresponding dimensionless quantity ($\Delta Z/Z$) carries no comparable empirical structure. The predictive information emerges only when the sequence is represented in terms of the intrinsic spectral boundary itself.

6. Spectral convergence and the limits of periodic differentiation

The preceding section showed that neighboring K-edge frequencies are not independent quantities but follow a highly constrained scaling law in which the fractional spacing between successive spectral ceilings converges toward a constant value. If this convergence reflects a deeper organizational principle, it should appear not only between neighboring atoms but also in the separation between entire chemical classes. The spectral framework reveals that this is indeed the case.

Periods and groups emerge in the $\nu_{\max}$ -based description as distinct branches occupying different regions of spectral space. Their separation provides a quantitative measure of periodic differentiation. When these branches remain widely separated, chemical classes are easily distinguishable. If the separation contracts, periodic distinctions become progressively compressed.

To quantify this effect, the periodic spectral bandwidth was defined as the separation between the alkali-metal (Group 1) and noble-gas (Group 18) boundaries in normalized spectral space.¹⁰ Across successive periods, this bandwidth follows a remarkably simple power-law relation with $\nu_{\max}$.

$$\Delta_p \propto \nu_{\max}^{-1.19 \pm 0.01} \quad (R^2 = 0.9995),$$

Over the experimentally accessible range, the bandwidth decreases by nearly two orders of magnitude, from the Ne–Na region to the Xe–Cs region. The contraction is systematic, monotonic, and exceptionally well described by a single scaling relation (Fig. 5, Ref. 10).

The significance of this result is that it measures the progressive compression of periodic structure itself. In light elements, the spectral space available for differentiation is comparatively large, producing well-separated chemical classes and strong contrasts between groups. As $\nu_{\max}$ increases, the available bandwidth shrinks, forcing the branches toward one another. The

periodic system therefore becomes increasingly crowded in spectral space even as atomic number continues to increase.

This convergence provides a natural explanation for one of the most persistent features of heavy-element chemistry: the remarkable similarity of the lanthanides and actinides. Extrapolation of the bandwidth relation places the f-block region within an exceptionally narrow spectral interval, where the separation between neighboring chemical trajectories approaches the limit of periodic resolution. In such a regime, distinct group identities become progressively less separable, producing the well-known chemical resemblance and classification ambiguities characteristic of these series.

Importantly, this result is not merely a description of f-block behavior. The contraction law is derived from the global spectral organization of the periodic system and predicts the f-block convergence without using lanthanide or actinide data in the fit. The observed similarity of these elements therefore emerges as a consequence of their position within the spectral coordinate system rather than as an isolated anomaly requiring a separate explanation. The lanthanide contraction, the actinide contraction, and the broader reduction in chemical differentiation at high atomic number appear as different manifestations of the same underlying spectral geometry.

Viewed together with the K-edge spacing law, a coherent picture emerges. At the local level, neighboring spectral ceilings converge toward a constant fractional spacing. At the global level, the bandwidth separating entire chemical classes contracts according to a power law. Both results point in the same direction: increasing spectral scale is accompanied by decreasing spectral capacity for differentiation. Atomic diversity is therefore not unbounded. It occupies a finite spectral manifold whose available resolution progressively decreases as ν_{max} increases.

The contrast with atomic-number representations is instructive. Atomic number preserves the sequential ordering of elements but contains no intrinsic measure of the separation between chemical classes. Equal increments in Z provide no indication of whether periodic differentiation is expanding, remaining constant, or contracting. Consequently, phenomena such as lanthanide similarity, actinide similarity, and the progressive compression of chemical behavior appear as additional observations requiring separate explanation. In the spectral space, by contrast, these effects emerge directly from a measurable contraction of periodic spectral bandwidth. The reduction in chemical differentiation is therefore not an independent phenomenon but a geometric consequence of the converging spectral structure itself, that was obscured when Z is used as the organizing coordinate.

The spectral spacing law and the spectral bandwidth contraction law reveal two complementary aspects of the same organization. Neighboring spectral ceilings become increasingly constrained relative to their local scale, while the spectral distance separating entire chemical classes simultaneously contracts. The first result establishes local convergence; the second establishes global convergence. Both indicate that atomic spectra evolve within a progressively compressed spectral geometry. The implication is that periodicity is not simply repeated indefinitely with increasing atomic number as Z -based representations suggest. Rather, it unfolds within a finite spectral manifold whose capacity for differentiation decreases systematically with scale. The same spectral organization that generates periods, groups, and chemical diversity also imposes

quantitative limits on their long-range separation. Spectral convergence therefore provides a natural endpoint to the progression developed throughout this work: from scaling, to branching, to periodic organization, to prediction, and finally to the finite resolution of periodic differentiation itself.

7. Spectral boundary control as a scale-independent organizing principle

The preceding sections established that $\nu_{\max}$, identified across atoms with the K-edge frequency, organizes multiple levels of spectral structure. At the atomic level, the total spectral bandwidth scales as a simple power law of $\nu_{\max}$ across more than ninety elements. At the periodic level, the periodic spectral bandwidth Δ_p likewise follows a power-law contraction with increasing $\nu_{\max}$. Thus, the same spectral boundary organizes both the internal extent of atomic spectra and the higher-order differentiation of the periodic system itself. These results suggest that $\nu_{\max}$ may represent a more general organizing principle than a parameter specific to K-edge spectroscopy.

A direct test of this possibility is whether the same organization persists across different spectral systems and across different frequency scales. The available data show that it does.⁹ Restricting the analysis to the ionization series of a single element eliminates variation in atomic number entirely. Across the ionization states Fe I–Fe XXIV, the relevant spectral boundary is no longer the atomic K-edge but the highest observed frequency of each ion. Despite this change in physical context, the spectral bandwidth remains tightly controlled by $\nu_{\max}$, following a near-unity scaling relation over several orders of magnitude in frequency (Fig. 3, Ref. 9). Ionization therefore does not generate a new global organization; instead, it rescales an existing spectral structure through changes in the system-defined upper boundary.

Hydrogen provides an even more stringent test because both multi-electron complexity and variation in atomic number are absent. Here, $\nu_{\max}$ is not associated with a K-edge and does not characterize the atom as a whole. Instead, each spectral family possesses its own upper frequency boundary, corresponding to its convergence limit. Analysis of the Lyman, Balmer, Paschen, and Brackett families shows that the family structure persists under normalization by these spectral boundaries and that the lower and upper limits of the families remain coupled through simple scaling relations (Fig. 6, Ref. 9). Once again, the relevant organizing variable is the spectral boundary itself rather than the specific physical mechanism that generates it.

These observations reveal a remarkable generalization. The same boundary-controlled organization appears across atoms, across periods, across ionization states within a single atom, and across spectral families within hydrogen. These systems span multiple hierarchical levels and many orders of magnitude in frequency, yet the same organizing principle persists: the accessible spectral structure is constrained by its upper frequency boundary. The empirical regularities therefore appear to depend not on the particular physical realization of the system but on the existence of a spectral boundary itself.

The contrast with atomic number is particularly revealing. Atomic number can order different elements, but it cannot organize ionization states within iron because Z remains fixed at 26. Nor

can it organize the spectral families of hydrogen because Z remains fixed at 1. Consequently, the scaling relations observed within Fe ionization states and within hydrogen families have no natural representation in a Z -based framework. These structures become visible only when the organizing coordinate belongs to the spectral domain itself.

This distinction highlights an important limitation of atomic-number indexing. Z functions as an external ordering parameter for elements that tracks increasing K-edge frequency, whereas $\nu_{\max}$ functions as an intrinsic coordinate defined by the spectral system under investigation, which in the case of neutral atoms, corresponds to the K-edge. Because it is system-specific rather than element-specific, $\nu_{\max}$ can be applied consistently across different levels of organization. The same coordinate that reveals the global organization of atomic spectra also describes spectral scaling within ions and within hydrogen families.

The empirical implication is that the organizing principle identified in this work is not the K-edge per se but spectral boundary control. Across every spectral system examined, the upper frequency limit acts as a global constraint on bandwidth, spacing, and internal organization. The recurrence of the same scaling behavior across such disparate systems suggests that boundary-controlled organization is a fundamental property of spectral systems, whereas atomic number remains restricted to its historical role as an ordering index for neutral elements alone.

III. Significance and Conclusions

The empirical findings synthesized in this work establish a progression from organization to explanation and finally to prediction and generalization. The $\nu_{\max}$ -based description reveals the branching architecture of atomic spectra, exposes the bifurcation structure through which periods emerge, explains anomalous elemental behaviors through measurable positions in spectral space, and uncovers scale-invariant laws capable of predicting previously unmeasured spectral boundaries. These diverse results arise from a common source: a low-dimensional spectral structure encoded in the system-specific spectral boundary $\nu_{\max}$.

Individually, each finding might be interpreted as a consequence of a particular dataset or analysis. Collectively, however, they reveal a coherent pattern that persists across atoms, ionization states, and hydrogen spectral families. The same coordinate organizes spectral scaling, periodic differentiation, predictive constraints, and convergence phenomena across systems that differ widely in physical realization and frequency scale. The significance of this pattern is not that Z produces incorrect results. Atomic number remains useful as an identifier of elemental order tracking K-edge frequencies. Rather, the empirical evidence shows that Z systematically obscures information that is directly accessible in the spectral domain. Branching structures collapse into sequential order. Bifurcation points become period labels. Scale-invariant spacing laws disappear into discrete increments. Converging spectral bandwidths become invisible. In some cases, the underlying organization is not lost entirely, but it appears in a fragmented and degraded form.

This observation clarifies the relationship between the two coordinates. Z and $\nu_{\max}$ (when represented by K-edge) are strongly correlated across the periodic table, and much of the empirical success of atomic-number representations derives from this correlation. However, the analyses presented here indicate that the deeper organizational structures belong to the spectral domain rather than to the integer index itself. To the extent that Z succeeds empirically, it does so because it tracks an underlying spectral organization that becomes more directly visible when represented through $\nu_{\max}$.

Perhaps the most significant result is that the same organizing principle persists across multiple scales. These systems differ in physical realization, span many orders of magnitude in frequency, and operate at different hierarchical levels, yet all exhibit the same boundary-controlled structure. The recurrence of this pattern suggests that the organizing principle identified here is not specific to atomic number, electron shells, or any particular atomic model. It appears instead to be a general property of bounded spectral systems.

The central conclusion is therefore empirical. Atomic spectra exhibit a low-dimensional organization that becomes visible when represented through their intrinsic spectral boundaries. This organization is measurable, predictive, scalable across multiple levels of structure, and recoverable directly from experimental data. Across atoms, ionization states, and hydrogen spectral families, the observed organization follows the system-specific spectral boundary $\nu_{\max}$. Atomic number participates only in the special case of neutral atoms, where $\nu_{\max}$ is represented by the atomic K-edge frequency; its apparent spectroscopic utility derives from inheriting the ordering of that particular realization of $\nu_{\max}$.

Statements and Declarations

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Data availability: The raw data supporting the findings of this study are publicly available at: For atomic spectral lines in the high frequency regime (X-ray) and maximum frequencies $\nu_{\max}$ (K-edge): <https://physics.nist.gov/PhysRefData/XrayTrans/Html/search.html> and <https://doi.org/10.1103/RevModPhys.75.35>

For atomic spectral lines in low frequency regime (UV-visible and infrared) and the spectral lines of iron ionized states: <https://doi.org/10.18434/T4W30F>

Competing Interests: The author is an Associate Professor of Nanotechnology at the University of Minnesota. This work was carried out independently and is not part of, or funded by the University. The author is also the founder and shareholder of *Spectral Dynamics, Inc.*, a startup company formed on January 6, 2026 after completion of the research reported here, which was developed over several years. The company has filed four patent applications related to potential

downstream applications of this work. The company had no role in the conception, execution, interpretation, or reporting of this work.

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